

0.1 A claim about how design and test choices interact

Joint optimization of test procedure and design parameters is the paper's central argument.

- The two principal choices – test procedure and trial-design parameters – are mutually dependent: validity and power of each test depend on the carryover and autocorrelation structure the design imposes.
- Treating the choices as independent optimizations, as the published literature typically does, produces locally optimal but jointly suboptimal combinations.
- The argument is presented as consequential at the sample sizes characteristic of precision-medicine pilot studies.

The paper's two-part argument establishes a gap in the literature and then fills it at the test-procedure axis.

- No joint design-by-test sensitivity study exists for the aggregated N-of-1 design class.
- The Monte Carlo experiment shows that linear-mixed with AR(1) dominates strict RM-ANOVA wherever residuals depart from compound symmetry.
- Strict RM-ANOVA is retained as a conservative stress-test baseline rather than a primary recommendation.

0.2 Why the question matters for chronic-condition pharmacology

Physiological autocorrelation in chronic-condition trials violates the sphericity assumption that strict RM-ANOVA requires.

- Chronic-condition response trajectories exhibit serial correlation driven by drug pharmacokinetics, natural-history drift, and measurement reactivity.
- Huynh-Feldt and Greenhouse-Geisser corrections recover nominal Type I only conditionally; the conventional threshold favors Greenhouse-Geisser when $\epsilon < 0.75$ (Blanca et al., 2023).
- The mixed-effects model with `corCAR1` avoids these corrections by modeling AR(1) structure directly.

Detecting the biomarker-treatment interaction is harder than detecting the average treatment effect (ATE) and is more sensitive to design choices.

- The PATH framework distinguishes effect-modeling from risk-modeling; the former requires more design-aware machinery.
- Within-participant contrasts feeding the interaction estimator are sensitive to trial-design parameters in ways the ATE analysis is not.
- Underpowered test procedures for the interaction will miss predictive-biomarker signal even at nominally adequate sample sizes.

66 **Cycle count and period length interact with carryover and test-procedure choice,**
67 **so their joint optimum cannot be found by optimizing each separately.**

- 68 • Additional cycles increase power for the biomarker-treatment interaction but with di-
69 minishing returns beyond the fourth or fifth cycle.
- 70 • Shorter periods produce more cycles but worsen on-drug/off-drug separation when car-
71 ryover is non-trivial.
- 72 • The joint optimum on the (cycles, period length) plane depends on carryover half-life
73 and test procedure together.

74 **0.3 What the published N-of-1 methodology literature does and does not** 75 **do**

76 **The majority of published N-of-1 analyses are conducted under a misspecified**
77 **residual covariance assumption.**

- 78 • Systematic reviews document that 83.8% of published N-of-1 studies ignore autocorre-
79 lation entirely.
- 80 • CONSORT extension standards codify reporting quality but do not address the analytic-
81 model decision.
- 82 • The analytic-method gap has been identified as the highest-priority methodological prob-
83 lem in the field.

84 **Existing test-procedure comparisons address parallel-group or cluster designs but**
85 **not the aggregated N-of-1 setting.**

- 86 • The mixed model for repeated measures (MMRM) consensus applies to parallel-group
87 longitudinal trials with monotone missingness, not N-of-1 designs.
- 88 • Only 6.6% of stepped-wedge cluster trials with fewer than six units use appropriate
89 small-sample corrections.
- 90 • No existing test-procedure comparison addresses the cross-over or N-of-1 setting directly.

91 **Design-parameter optimization has been studied in isolation from test-procedure**
92 **choice, leaving the joint question unaddressed.**

- 93 • Hendrickson et al. (2020) is the only paper comparing full design families for biomarker-
94 interaction power, but holds period length fixed and uses a single test procedure.
- 95 • Power calculations, autocorrelation effects, and cross-over design literature each address
96 one axis without crossing them.
- 97 • No published study crosses test procedure against the design-parameter grid under sim-
98 ulation for the aggregated N-of-1 class.

99 **No published study crosses test procedure against a design-parameter grid for**
100 **the aggregated N-of-1 design class.**

- 101 • The gap spans test-procedure comparison (strict RM-ANOVA, MMRM, GEE, mixed-
102 effects with `corCAR1`) crossed against cycles, period lengths, asymmetry, run-in, and

103 run-out.

- 104 • The present paper addresses the gap on the test-procedure axis.
- 105 • The design-parameter axis is deferred to a companion paper.

106 0.4 Run-in design and the placebo-selection controversy

107 **The Hendrickson hybrid design avoids placebo-run-in selection bias by using open-**
108 **label stabilization, at some power cost.**

- 109 • Response-based exclusion during a placebo run-in introduces selection bias that damages
110 external and potentially internal validity (Berger et al., 2003).
- 111 • The Hendrickson open-label stabilization phase avoids this bias but forgoes the power
112 that aggressive response-based selection would supply.
- 113 • The run-in choice is held fixed in the present paper; varying it is deferred to the com-
114 panion design-sensitivity paper.

115 0.5 What this paper contributes

116 **The first contribution is a closed-form strict RM-ANOVA F -statistic for the**
117 **biomarker-treatment interaction and its relation to the linear-mixed Wald test.**

- 118 • The derivation specializes the balanced split-plot model to the biomarker-stratified case
119 under the sphericity assumption.
- 120 • The conditions under which the two tests agree and diverge are identified explicitly.
- 121 • The derivation extends standard cross-over methodology to the biomarker-stratified
122 case.

123 **The second contribution is specification of the cycle-by-period design grid cali-**
124 **brated to the prazosin/PTSD reference application.**

- 125 • The grid varies cycle count, on-drug and off-drug period lengths, and on/off symmetry
126 around the Hendrickson hybrid reference.
- 127 • Calibration is to the prazosin/PTSD application that motivates the pmsimstats pro-
128 gram.
- 129 • The empirical sweep across this grid is deferred to a companion paper.

130 **The third contribution is a Monte Carlo simulation study crossing test procedures**
131 **against the design grid under the ADEMP framework.**

- 132 • The simulation runs on the pmsimstats package data-generating process under the mean-
133 moderation architecture, exercised through `generateData()` and `lme_analysis()`.
- 134 • Five test procedures, including a continuous-biomarker contrast that isolates the di-
135 chotomization cost, are crossed against biomarker-moderation coefficient and sample
136 size in a 40-cell factorial.
- 137 • Results are reported following the Morris et al. (2019) ADEMP discipline including
138 Monte Carlo standard errors (MCSE), bias, and interval coverage.

139 **The fourth contribution is a design-and-test recommendation table indexed by**
140 **carryover half-life and target effect size.**

- 141 • The recommendation is presented as preferred (cycle count, period length, test proce-
142 dure) combinations rather than a single best design.
- 143 • The recommendation is narrowed to the test-procedure axis at the reference design in
144 the present paper.
- 145 • The joint cycle-by-period optimization is reserved for the companion paper.

146 **The paper’s organization maps each section to one of the four contributions.**

- 147 • Sections 2 and 3 cover the closed-form derivation and the design-grid specification re-
148 spectively.
- 149 • Sections 4 and 5 present the simulation design and results.
- 150 • Sections 6 and 7 synthesize the recommendation and discuss implications.

151 **1 Closed-form strict RM-ANOVA test of the biomarker-** 152 **treatment interaction**

153 **The section derives the strict RM-ANOVA closed form and identifies three find-**
154 **ings that anchor the rest of the paper.**

- 155 • The derivation is confined to the strict classical compound-symmetric form; modern
156 extensions appear only for context.
- 157 • The simplest design supporting the test has two biomarker strata, two treatment phases,
158 and T within-phase timepoints.
- 159 • Three findings from the derivation – equivalence to a two-sample t -test, a closed-form
160 power formula, and five data-generating process (DGP) violations – anchor the simula-
161 tion study.

162 **1.1 Notation and design**

163 **The notation section establishes a restricted design that is deliberately simpler**
164 **than the pmsimstats DGP.**

- 165 • The design is the simplest that supports a biomarker-treatment interaction test in a
166 within-subject setting.
- 167 • The gap between this restricted design and the pmsimstats DGP is addressed explicitly
168 in Section 2.4.
- 169 • All departures from the DGP are flagged as sources of conservative bias in the strict
170 RM-ANOVA operating characteristics.

171 **The split-plot structure arises because the biomarker stratum is between-subjects**
172 **while treatment phase and timepoints are within-subjects.**

- Strict RM-ANOVA requires categorical predictors; dichotomizing the continuous biomarker at a cut-point is the price of admission.
- Each subject is nested in exactly one biomarker stratum but passes through both treatment phases.
- The design is therefore a canonical split-plot repeated-measures design.

The classical split-plot model encodes the biomarker-treatment interaction as the $(\alpha\beta)_{ij}$ fixed effect under compound symmetry.

- Sum-to-zero constraints are imposed on all fixed effects; the random subject effect is nested in biomarker stratum.
- The sphericity (compound symmetry) condition requires equal within-subject covariance across all observation pairs.
- The biomarker-treatment interaction $(\alpha\beta)_{ij}$ and its time-resolved form $(\alpha\beta\tau)_{ijk}$ are the primary targets.

1.2 Sums of squares and the interaction F -statistic

The total sum of squares partitions into between-subjects and within-subjects components that are orthogonal under balance.

- The between-subjects component captures biomarker-stratum effects and subject-within-stratum variation.
- The within-subjects component captures treatment-phase effects, the biomarker-treatment interaction, and within-subjects error.
- Orthogonality under balance ensures the interaction F -ratio uses the correct within-subjects error term.

The interaction F -ratio uses the within-subjects error $SS_{B \times S(A)}$ as its denominator, with degrees of freedom $a(n-1)(b-1)$.

- SS_{AB} has $df_{AB} = (a-1)(b-1)$ degrees of freedom; the within-subjects error has $df_{B \times S(A)} = a(n-1)(b-1)$.
- The within-subjects error term is the appropriate denominator for the treatment effect and its interaction with the between-subjects factor.
- The F -ratio $F_{AB} = MS_{AB}/MS_{B \times S(A)}$ follows a central F under H_0 and sphericity.

1.2.1 The 2×2 split-plot case

Specializing to $a = b = 2$ with $T = 1$ reduces the interaction to the difference of within-subject treatment differences across biomarker strata.

- Setting $T = 1$ corresponds to pre-averaging within-phase observations before computing the interaction.
- The sample interaction contrast $\hat{L}$ is the difference of within-subject treatment differences across strata.

- This specialization yields the cleanest closed-form expression for the interaction F -statistic.

Under sphericity and balance in the 2×2 case, the interaction F -statistic reduces to $nT\hat{L}^2/(4\hat{\sigma}_e^2)$ with degrees of freedom $(1, 2(n-1))$.

- The interaction sum of squares is $SS_{AB} = nT\hat{L}^2/4$; with $df_{AB} = 1$ this equals the interaction mean square.
- Under sphericity, $MS_{B \times S(A)} = \hat{\sigma}_e^2$, yielding the closed form directly.
- The error degrees of freedom are $2(n-1)$, reflecting one degree of freedom per subject in each of the two strata.

Equation ?? is the paper's strict RM-ANOVA F reference expression and is named to distinguish it from the mixed-effects and GEE Wald statistics.

- The expression applies to the $2 \times 2 \times T$ split-plot design and is the textbook compound-symmetric form.
- It serves as the analytic anchor for the power formula in Section 2.3.1 and for comparison with simulation results in Section 5.
- The name 'strict RM-ANOVA F ' is used throughout to mark the distinction from Wald statistics that do not require sphericity.

1.2.2 Equivalence to a within-subject t -test

The strict RM-ANOVA F -test for the biomarker-treatment interaction is equivalent to a two-sample t -test on within-subject treatment differences.

- The within-subject treatment difference Δ_{il} cancels the random subject effect, leaving a pure within-subject contrast.
- $\hat{L} = \bar{\Delta}_1 - \bar{\Delta}_2$ is the difference of stratum-mean treatment differences.
- The equivalence $t_{AB}^2 = F_{AB}$ is well-known in the split-plot literature and makes the test's assumptions unusually transparent.

$t_{AB}^2 = F_{AB}$ recovers the interaction F -statistic, confirming that the strict RM-ANOVA test is a two-sample comparison of within-subject treatment effects.

- The identity holds under sphericity because the random subject effect cancels in within-subject differences.
- The test reduces to asking whether the within-subject treatment effect differs between biomarker strata.
- Transparency about this reduction is analytically useful: anything that biases within-subject differences distorts the test in a predictable direction.

The transparent reduction to a two-sample t -test will be used in Section 5 to interpret the simulation results.

- The reduction is exploited in Section 5 to connect strict RM-ANOVA results to the mixed-effects findings.

- Biases and inflations in within-subject differences produce predictable distortions in the test's operating characteristics.
- The intuitive appeal of the reduction makes the test's limitations equally transparent.

The general $a \times b$ case and the time-resolved three-way interaction extend equation ?? with degrees of freedom scaled by $(a - 1)(b - 1)$.

- For arbitrary a and b , the interaction F -statistic uses $df_1 = (a - 1)(b - 1)$ and $df_2 = a(n - 1)(b - 1)$.
- The three-way interaction $(\alpha\beta\tau)_{ijk}$ tests whether the biomarker modifies the treatment effect across time.
- The corresponding degrees of freedom are $(a - 1)(b - 1)(T - 1)$ and $a(n - 1)(b - 1)(T - 1)$.

1.3 Relation to the linear-mixed Wald test

The strict RM-ANOVA F and the linear-mixed Wald t are asymptotically equivalent only under three conditions that the pmsimstats DGP does not satisfy simultaneously.

- Equivalence requires dichotomized biomarker, compound-symmetric within-subject covariance, and binary Dbc.
- When any condition fails, the mixed-model test extracts more information via continuous biomarker, AR(1) modeling, or continuous exposure-decay.
- In the pmsimstats setting, all three conditions fail, so the mixed-model test dominates strictly.

Sphericity violations inflate Type I error; Greenhouse-Geisser and Huynh-Feldt corrections recover nominal control by penalising degrees of freedom.

- Mauchly's test detects compound-symmetry failure but is not informative about direction; Box's ϵ is a scalar summary with $\epsilon = 1$ at perfect sphericity.
- The Greenhouse-Geisser correction adjusts both numerator and denominator degrees of freedom by $\hat{\epsilon}$; Huynh-Feldt is less conservative and is recommended when $\hat{\epsilon} > 0.75$.
- The conventional rule recommends Greenhouse-Geisser when $\hat{\epsilon} < 0.75$ to avoid substantial Type I error inflation.

Under the pmsimstats AR(1) DGP, a mixed model with no residual-correlation structure inflates Type I error to 13–17%; corCAR1 restores nominal control, and the pre-averaged strict RM-ANOVA arm is sphericity-robust.

- AR(1) within-factor correlation is not compound-symmetric; Box's ϵ is strictly below 1.
- Greenhouse-Geisser correction recovers nominal Type I error but substantially attenuates power by penalising degrees of freedom.
- The corCAR1 mixed-model analysis models AR(1) structure directly and does not pay the degrees-of-freedom penalty.

The strict RM-ANOVA test is a conservative reduction rather than a competing

283 **alternative; option (c) – reporting both with explicit power-loss tabulation – is**
284 **operationalized in the simulation study.**

- 285 • Three defensible responses to a reviewer requesting RM-ANOVA results are identified,
286 ranging from reframing the LME analysis to reporting both with power-loss tables.
- 287 • Option (c), reporting both procedures with explicit power-loss tabulation, is the strategy
288 adopted in Sections 4–5.
- 289 • This strategy is the most transparent and the most informative for the reviewer.

290 1.3.1 Power calculation under the closed form

291 **The non-centrality parameter $\lambda = nT\delta^2/(4\sigma_e^2)$ provides the analytic power anchor**
292 **for the simulation study.**

- 293 • Under the alternative $H_1 : \hat{L} = \delta$, the F -statistic follows a non-central F distribution.
- 294 • The non-centrality parameter scales linearly with n and T and quadratically with δ .
- 295 • The formula inverts to a sample-size formula $n = 4\sigma_e^2\lambda/(T\delta^2)$ for power planning.

296 **All five assumptions of the closed-form power formula are violated to varying**
297 **degrees in the pmsimstats DGP, so the formula provides an upper bound rather**
298 **than a calibrated prediction.**

- 299 • The five assumptions are: sphericity, categorical biomarker with equal stratum sizes, no
300 carryover, no within-treatment relapse, and i.i.d. within-phase observations.
- 301 • Violations of sphericity and i.i.d. within-phase observations are two faces of the same
302 AR(1) departure from compound symmetry.
- 303 • The analytic power formula is best understood as an upper bound on power available
304 from strict RM-ANOVA under the pmsimstats DGP.

305 **The five assumptions are not independent: sphericity and i.i.d. within-phase ob-**
306 **servations are two faces of the same AR(1) departure from compound symmetry.**

- 307 • Under AR(1), compound-symmetry failure drives both Type I error inflation and within-
308 phase autocorrelation simultaneously.
- 309 • The analytic power formula is therefore a joint upper bound, not a calibrated predictor
310 for either failure mode in isolation.
- 311 • This dependence motivates the Monte Carlo validation in Sections 4–5 rather than
312 reliance on the closed form alone.

313 1.3.2 Limitations of the strict classical formulation

314 **The five assumptions of the strict classical formulation are empirically problematic**
315 **in the pmsimstats setting; their severity is presented in order.**

- 316 • Closed-form simplicity is bought at the price of categorical biomarker, sphericity, no
317 carryover model, balanced design, and equal time spacing.

- Each assumption is violated by the pmsimstats DGP to a degree that favors the linear-mixed analysis.
- The most severe violations – categorical biomarker and no carryover – together account for a 30–50% attenuation of the interaction estimator.

The five limitations motivate the primary-and-sensitivity reporting strategy; the categorical-biomarker and no-carryover violations are the most consequential.

- The categorical-biomarker and no-carryover violations together account for a 30–50% attenuation of the interaction estimator relative to the linear-mixed analysis.
- The linear-mixed analysis with `corCAR1` is retained as primary; the strict RM-ANOVA result is reported as a sensitivity comparator.
- The Section 6 recommendation operationalizes this strategy for typical chronic-condition N-of-1 scenarios.

2 Cycle-by-period design grid

The cycle-by-period grid is specified here but its empirical sweep is deferred; the Section 6 recommendation is correspondingly narrowed to the test-procedure axis.

- The design grid varies cycle count, on-drug and off-drug period lengths, and on/off symmetry around the Hendrickson hybrid reference.
- The present paper’s scope is narrowed to the test-procedure axis at the prazosin/PTSD reference design.
- The joint (cycle count, period length, test procedure) optimum is deferred to the follow-up paper and recorded as a portfolio item.

3 Simulation design

The ADEMP framework structures the simulation program around the biomarker-treatment interaction estimand under five test procedures.

- The five procedures are strict RM-ANOVA (dichotomized biomarker), a continuous-biomarker within-subject contrast, linear-mixed with `corCAR1`, GEE with naive sandwich, and GEE with Mancl-DeRouen small-sample correction.
- The continuous-biomarker contrast separates the cost of dichotomization from the cost of the test procedure proper; the four continuous-biomarker arms share a common interaction estimand.
- Primary outputs are power and Type I error at $\alpha = 0.05$; secondary estimands (bias, empirical and model SE, 95% CI coverage) are reported for the four coefficient-bearing arms.
- All estimates are reported with MCSE following the Morris et al. (2019) discipline.

353 **MCSE for power and Type I error follow the binomial-proportion form and are**
354 **reported alongside every estimate.**

- 355 • Power and Type I error MCSE use $\sqrt{\hat{\pi}(1 - \hat{\pi})/n_{\text{reps}}}$.
- 356 • Bias and empirical SE of $\hat{\beta}_{bm:D}$ use the standard within-replicate mean estimator.
- 357 • Reporting MCSE alongside every estimate is the Morris et al. discipline adopted project-
- 358 wide.

359 **The full ADEMP pre-registration fixes three studies before production runs com-**
360 **mit, providing a pre-specified analytic plan.**

- 361 • Study 1 is the test-procedure comparison at fixed design (27 cells); Studies 2 and 3 cover
- 362 the cycle-by-period grid and joint heatmap (240 cells each).
- 363 • Pre-registration at 00-ademp-pre-registration.md locks the factorial before produc-
- 364 tion runs commit.
- 365 • The present paper reports only Study 1; Studies 2 and 3 are deferred to the companion
- 366 paper.

367 **The factorial crosses five procedures by two β_{bm} values by four N values; the inline**
368 **Mancl-DeRouen SE is benchmarked against `geesmv::GEE.var.md`.**

- 369 • The simulation crosses five procedures by $\beta_{bm} \in \{0, 0.45\}$ by $N \in \{25, 35, 70, 100\}$, with
- 370 1000 replicates per power cell and 2000 per Type I cell.
- 371 • The Mancl-DeRouen correction inflates each cluster's residual via $(I - H_{ii})^{-1}r_i$ in the
- 372 sandwich meat.
- 373 • The inline SE agrees with `geesmv::GEE.var.md` to within roughly 2.5% (mean ratio
- 374 1.02 over 25 probe replicates), so the correction magnitude is externally validated.

375 4 Results

376 **The 40-cell factorial completed in 587 s with 100% convergence across 60,000 fits;**
377 **MCSE is approximately 0.016 for power and 0.005 for Type I error.**

- 378 • Execution used 7-core `parallel::mclapply` with L'Ecuyer-CMRG streams; conver-
- 379 gence was 100% in every cell.
- 380 • Per-replicate output and the ADEMP summary are archived at `analysis/data/quick-sim/`.
- 381 • MCSE on power near 0.5 is 0.016 (1000 reps); MCSE on Type I error near 0.05 is 0.005
- 382 (2000 reps), providing adequate resolution to distinguish the procedures.

383 4.1 Test-procedure comparison at fixed design

384 **Strict RM-ANOVA targets a dichotomized estimand; the continuous-biomarker**
385 **contrast isolates the cost of dichotomization from the cost of the test procedure**
386 **proper.**

- Strict RM-ANOVA operates on a median-dichotomized biomarker, pre-averaged to one value per phase; its estimand is on the dichotomized scale.
- The continuous-biomarker contrast uses the same pre-averaging but the continuous biomarker, so it shares the estimand of the linear-mixed and GEE arms.
- The three-rung ladder (RM-ANOVA, contrast, linear-mixed) decomposes the gap between classical RM-ANOVA and the mixed model into a dichotomization component and a test-procedure component.

Most of the gap between strict RM-ANOVA and the mixed model is the cost of dichotomization, not of the test procedure: the continuous-biomarker contrast recovers four-fifths of it.

- At $N = 25$, replacing the dichotomized biomarker with the continuous one (RM-ANOVA $\rightarrow$ contrast) raises power from 0.358 to 0.538, a gain of 0.180.
- Moving from the pre-averaged contrast to the full linear-mixed model adds only 0.027 (0.538 $\rightarrow$ 0.565); the test-procedure component is small.
- The same pattern holds across N : dichotomization costs 0.16–0.21 of power, while the longitudinal test procedure adds at most 0.04.

Naive GEE inflates Type I error across the whole sample-size range, not only at small N ; the Mancl-DeRouen correction restores nominal control at a moderate power cost.

- Naive GEE rejects under the null at 0.097, 0.086, 0.079, and 0.074 for $N = 25, 35, 70, 100$ – between $1.5\times$ and $1.9\times$ nominal throughout.
- The Mancl-DeRouen correction brings the rate to 0.064, 0.053, 0.063, and 0.059, within or close to MCSE of the nominal 0.05 at every N .
- The correction’s power cost is moderate (about 0.10 at $N = 25$, 0.07 at $N = 35$) and the corrected procedure no longer over-rejects.

Standard-error calibration separates the four coefficient-bearing arms: the contrast and Mancl-DeRouen GEE are well-calibrated, naive GEE under-covers, and the linear-mixed model over-covers.

- The continuous-biomarker contrast and Mancl-DeRouen GEE attain 93–95% coverage with model/empirical SE ratios near 1.
- Naive GEE under-covers at small N (90.0% at $N = 25$) because its model SE is 9–12% too small.
- The linear-mixed model over-covers (97.0–97.7%) because its model SE is 11–16% larger than the empirical SE, the calibration signature of its conservatism.

All four coefficient-bearing arms are approximately unbiased; the linear-mixed conservatism is a variance-calibration effect, not a point-estimate effect, and a Kenward-Roger correction would address it.

- Across all cells, the relative bias of the interaction coefficient is below 4% of the estimand for every coefficient-bearing arm.

- The linear-mixed Type I error is 0.021 to 0.034, below nominal, consistent with `corCAR1` degrees-of-freedom accounting under $T = 8$ timepoints.
- A Kenward-Roger or Satterthwaite small-sample correction would address the conservatism and is recorded as a follow-up item.

5 Design-and-test recommendation

The recommendation is presented as a table of preferred (procedure, scenario) combinations at the reference design rather than as a single best procedure.

- Cycle count and period length are held at reference values; joint optimization over those axes is deferred to the follow-up paper.
- The preferred procedure depends on features of the application – sample size, tolerance for Type I inflation, and reviewer requirements – that the analyst must determine.
- The first-order recommendation is to keep the biomarker continuous; the choice among continuous-biomarker procedures is second-order.

Keep the biomarker continuous; among continuous-biomarker procedures the linear-mixed model is the preferred default, and naive GEE should not be used without a small-sample correction at any N examined.

- The first-order decision is to retain the continuous biomarker; dichotomization forfeits 0.08–0.21 of power across the grid.
- Among continuous-biomarker procedures, linear-mixed with `corCAR1` has the highest power within nominal Type I; the contrast is a close, transparent alternative.
- Naive GEE is anti-conservative at every N examined and should be replaced by the Mancl-DeRouen correction where a sandwich estimator is required.

5.1 What the Monte Carlo study settles

The simulation settles four substantive questions; the inline Mancl-DeRouen SE is validated against `geesmv::GEE.var.md` to within 2.5%.

- The 40-cell factorial completed in 587 s with 100% convergence across 60,000 fits; MCSE is 0.016 for power near 0.5 and 0.005 for Type I error near 0.05.
- The inline Mancl-DeRouen implementation produces identical point estimates and larger standard errors than the naive sandwich.
- Its SE for the interaction coefficient matches `geesmv::GEE.var.md` to within roughly 2.5% over 25 probe replicates, so the correction magnitude is externally validated.

The first resolved question is that the classical-versus-mixed power gap is mostly the cost of dichotomization: the continuous-biomarker contrast recovers about four-fifths of it.

- At $N = 25$, dichotomization (RM-ANOVA $\rightarrow$ contrast) costs 0.180 of power; the test procedure (contrast $\rightarrow$ linear-mixed) adds 0.027.

- The decomposition is stable across N : dichotomization costs 0.08–0.21, the test procedure at most 0.04.
- A classical, sphericity-free, AR(1)-free contrast on the continuous biomarker therefore captures most of the available power.

The second resolved question is that naive GEE inflates Type I error across the entire sample-size range, and Mancl-DeRouen restores nominal control.

- Naive GEE rejects at 0.097, 0.086, 0.079, 0.074 for $N = 25, 35, 70, 100$ – still about $1.5\times$ nominal at $N = 100$.
- The Mancl-DeRouen correction brings the rate to 0.064, 0.053, 0.063, 0.059, within or near MCSE of nominal at every N .
- The power cost is moderate (about 0.10 at $N = 25$, 0.07 at $N = 35$).

The third resolved question is standard-error calibration: the contrast and Mancl-DeRouen GEE are well-calibrated, naive GEE under-covers, and the linear-mixed model over-covers.

- Contrast and Mancl-DeRouen GEE attain 93–95% coverage with model/empirical SE ratios near 1.
- Naive GEE under-covers (90.0% at $N = 25$) because its model SE is 9–12% too small.
- The linear-mixed model over-covers (97.0–97.7%) because its model SE exceeds the empirical SE by 11–16%.

The fourth resolved question is that the linear-mixed conservatism is a degrees-of-freedom artifact a Kenward-Roger correction would remove.

- LME Type I error is 0.021 to 0.034, conservative rather than nominal, consistent with `corCAR1` degrees-of-freedom accounting under $T = 8$ timepoints.
- The conservatism shows up as over-coverage and a model/empirical SE ratio above one, not as point-estimate bias.
- A Kenward-Roger or Satterthwaite correction would recover the small power increment the conservatism forgoes.

6 Conclusions

The biomarker should be kept continuous: dichotomization, not the test procedure, drives most of the classical-versus-mixed power gap; naive GEE is disqualified without a small-sample correction.

- Dichotomizing the biomarker for strict RM-ANOVA forfeits 0.08–0.21 of power across N ; the test procedure proper adds at most 0.04.
- Among continuous-biomarker procedures the linear-mixed model has the highest power within nominal Type I; the contrast is a close, transparent alternative.
- Naive GEE inflates Type I error to 1.5 – $1.9\times$ nominal across all N ; the Mancl-DeRouen correction restores nominal control at a moderate (about 10-percentage-point) power

499 cost.

500 **The closed-form reduction to a two-sample t -test makes the strict RM-ANOVA**
501 **transparent; in this study its limitation is dichotomization, with sphericity rele-**
502 **vant only to the un-pre-averaged classical analysis.**

- 503 • Under sphericity and balance, the biomarker-treatment interaction F -test reduces to a
504 two-sample t -test on within-subject treatment differences.
- 505 • The pre-averaged-to-one-value-per-phase form evaluated here is sphericity-robust by con-
506 struction and holds nominal Type I error, so its power loss is attributable to dichotomiza-
507 tion, not to sphericity.
- 508 • Sphericity violation under AR(1) bites only when the full multi-timepoint series is ana-
509 lyzed by classical RM-ANOVA without a degrees-of-freedom correction; the linear-mixed
510 model with `corCAR1` sidesteps both issues.

511 **The cycle-by-period design grid is deferred to a follow-up paper; infrastructure**
512 **and design specification are in place and the scope limitation is explicitly ac-**
513 **knowledged.**

- 514 • The simulation infrastructure for the cycle-by-period sweep is committed at
515 `analysis/scripts/test-procedure-design-sensitivity/`.
- 516 • The design specification is at `docs/27-cycle-period-design-sweep-plan.md`.
- 517 • The follow-up paper will deliver the joint recommendation table that the present paper's
518 Section 6 narrows to the test-procedure-only axis.

519 7 Data availability statement

520 **Data and code are openly available in the pmsimstats-ng repository at paths listed**
521 **in the Reproducibility section.**

- 522 • Per-replicate Monte Carlo outputs and ADEMP cell-level summaries are versioned under
523 `analysis/data/`.
- 524 • Simulation drivers are at `analysis/scripts/`.
- 525 • Exact paths relevant to this manuscript are listed in the Reproducibility section.